

# Luigui Gallardo-Becerra

BIOINFORMATICIAN | MOLECULAR BIOLOGIST

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*Experienced Bioinformatician and Molecular Biologist seeking new challenges to apply and expand my expertise in computational biology, data science, and molecular research. Over the past several years, I have contributed to diverse research projects utilizing modern genomic technologies (NGS), advanced data analysis, and data science methodologies to generate novel biological insights that have resulted in high-impact peer-reviewed scientific publications.*

## Research Experience

### Graduate Research Assistant (Bioinformatics & Molecular Biology)

*Institute of Biotechnology, UNAM*

*January 2019 - July 2025*

Across several interdisciplinary research projects, I contributed to data acquisition, computational analysis, and dissemination of scientific results. My responsibilities included:

**Project conceptualization:** Assisted in defining research goals, experimental design, analytical workflows, and computational strategies to address multi-omics biological questions.

**HPC system administration:** Maintained laboratory computational infrastructure, including Linux installation, software configuration, performance optimization, and data security practices.

**Reproducible workflow engineering:** Built robust, scalable, and reproducible pipelines using Snakemake, Python, Bash, and Rabix Composer to automate NGS and metagenomic analyses.

**Custom tool development:** Implemented specialized algorithms and utilities in Python, R, Bash, and Perl to support complex multi-step analyses.

**Quantitative data visualization:** Designed comprehensive visualizations and figure panels using ggplot2, seaborn, matplotlib, and Jupyter Notebooks.

**Computational best practices:** Maintained version control through GitHub repositories and incorporated CI/CD testing to ensure workflow reliability.

**Scientific reporting:** Prepared internal research reports, contributed to manuscript writing, and provided analytical support for high-impact publications.

**Presentation & outreach:** Delivered oral and poster presentations at national and international conferences and participated in science communication initiatives.

## Education

### National Autonomous University of Mexico (UNAM)

PH.D. IN COMPUTATIONAL BIOLOGY

*Mexico City, Mexico*

*January 2019 - July 2025*

### National Autonomous University of Mexico (UNAM)

MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY

*Mexico City, Mexico*

*August 2016 - January 2019*

### University of Guadalajara (UDG)

BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY

*Guadalajara, Mexico*

*August 2012 - January 2016*

## Teaching Experience

- Guest lecturer at Universidad de La Sabana (Chia, Colombia). December 2023.
- Organizer and instructor of the “Bioinfo del Norte” bioinformatics and metagenomics training event, CIAD, Hermosillo, Sonora. September 2024.

- Microbiota Symposium from Theory to Clinical Practice, Colegio de Profesionales de la Nutrición de Querétaro y el Bajío. June 2021.
- Bioinformatics and Reproducible Research Workshop, Postgraduate Program in Biological Sciences, UNAM. October 2019.
- Guest lecturer at Center of Genomic Sciences, UNAM. August 2018.

## Directed Theses

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Bachelor's Thesis: Itzel Abigail Hernández Reyna, Bachelor's in Nutrition (Universidad Autónoma del Estado de Morelos), December 2019. "Impact of bacteriophages associated with childhood obesity on the intestinal metagenome."

## Soft Skills

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Adaptable and quick to learn new technologies, analytical methods, and computational tools.

Strong problem-solving skills with the ability to manage multiple complex projects while maintaining high-quality output.

Effective communicator with experience collaborating in interdisciplinary teams as well as working independently.

Skilled in technical operations including server maintenance, Git/GitHub workflows, and development of automated scripts for research tasks.

## Programming Languages, Tools, Bioinformatics & Molecular Biology Skills

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### Specialized Bioinformatics Skills:

Genome and transcriptome assembly & annotation; differential expression analysis; taxonomic profiling; functional & pathway analysis; ETL automation for multi-omics data

### Bioinformatics Tools & Methods:

Snakemake, Nextflow, QIIME2, Trinity, SPAdes, Bowtie2, BWA, HISAT2, samtools, fastp, Kraken2

### NGS & Multi-omics Analysis:

RNA-seq, WGS, 16S rRNA profiling, metagenomics, metatranscriptomics, viromics

### Molecular Biology Skills:

DNA/RNA extraction, PCR/qPCR, gel electrophoresis, sample QC, NGS library preparation, bacterial culture & handling, plasmid preparation, sterile technique (BSL-2), and collaboration with wet-lab teams

### Programming Languages:

Python, R, Bash, SQL, C#, JavaScript, TypeScript, HTML/CSS

### Frameworks & Libraries:

React, Django, ASP.NET Core, Shiny

### Computing & DevOps:

Linux, High-Performance Computing, Docker, Git/GitHub, CI/CD workflows, Conda

### Data Science & Visualization:

ggplot2, seaborn, matplotlib, Jupyter Notebooks, RStudio

### Databases:

MySQL, PostgreSQL, MongoDB

## Languages

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English – Full professional proficiency

Spanish – Native

## Publications

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1. Jatuyosporn T, Laohawutthichai P, Cornejo-Granados F, **Gallardo-Becerra L**, Tassanakajon A, Hurtado-Ramírez JM, Ochoa-Leyva A, Krusong K. *De novo* transcriptome assembly and annotation of *Penaeus monodon* hemocytes under WSSV infection and STAT knockdown. *Sci Data*. 2026.
2. **Gallardo-Becerra L**, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, Sánchez-López F, Manzo R, Corzo G, Espino-Solis GP, Canizales-Quinteros S, Ochoa-Leyva A. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. *Microb Ecol*. 2025 Nov 28.
3. Cornejo-Granados F, **Gallardo-Becerra L**, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huizar A, Cervantes-Echeverría M, Sotelo-Mundo RR, Ochoa-Leyva A. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. *Animal Microbiome*. 2025 May 22;7(1):50.
4. Manzo R, **Gallardo-Becerra L**, Díaz de León-Guerrero S, Villaseñor T, Cornejo-Granados F, Salazar-León J, Ochoa-Leyva A, Pedraza-Alva G, Pérez-Martínez L. Environmental Enrichment Prevents Gut Dysbiosis Progression and Enhances Glucose Metabolism in High-Fat Diet-Induced Obese Mice. *International Journal of Molecular Sciences*. 2024 Jan;25(13):6904.
5. Jatuyosporn T, Laohawutthichai P, Ochoa Romo JP, **Gallardo-Becerra L**, Sánchez Lopez F, Tassanakajon A, Ochoa-Leyva A, Krusong K. White spot syndrome virus impact on the expression of immune genes and gut microbiome of black tiger shrimp *Penaeus monodon*. *Sci Rep*. 2023 Jan 18;13(1):996.
6. **Gallardo-Becerra L**, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in Searching Antimicrobial Peptides (AMPs) Produced by the Microbiota. *Microb Ecol*. 2023 Dec 1;87(1):8.
7. Chino de la Cruz CM, Cornejo-Granados F, **Gallardo-Becerra L**, Rodríguez-Alegría ME, Ochoa-Leyva A, López Munguía A. Complete genome sequence and characterization of a novel *Enterococcus faecium* with probiotic potential isolated from the gut of *Litopenaeus vannamei*. *Microbial Genomics*. 2023;9(3):000938.
8. Cervantes-Echeverría M, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. The Two-Faced Role of crAssphage Subfamilies in Obesity and Metabolic Syndrome: Between Good and Evil. *Genes*. 2023 Jan;14(1):139.
9. Palomino-Hermosillo YA, Berumen-Varela G, Ochoa-Jiménez VA, Balois-Morales R, Jiménez-Zurita JO, Bautista-Rosales PU, Martínez-González ME, López-Guzmán GG, Cortés-Cruz MA, Guzmán LF, Cornejo-Granados F, **Gallardo-Becerra L**, Ochoa-Leyva A, Alia-Tejacal I. Transcriptome Analysis of Soursop (*Annona muricata* L.) Fruit under Postharvest Storage Identifies Genes Families Involved in Ripening. *Plants (Basel)*. 2022 Jul 7;11(14):1798.
10. Ochoa-Romo JP, Cornejo-Granados F, Lopez-Zavala AA, Viana MT, Sánchez F, **Gallardo-Becerra L**, Luque-Villegas M, Valdez-López Y, Sotelo-Mundo RR, Cota-Huizar A, López-Munguía A, Ochoa-Leyva A. Agavin induces beneficial microbes in the shrimp microbiota under farming conditions. *Sci Rep*. 2022 Apr 16;12(1):6392.
11. Bikel S, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. Protocol for the isolation, sequencing, and analysis of the gut phageome from human fecal samples. *STAR Protocols*. 2022 Mar 18;3(1):101170.
12. Bikel S, López-Leal G, Cornejo-Granados F, **Gallardo-Becerra L**, García-López R, Sánchez F, Equihua-Medina E, Ochoa-Romo JP, López-Contreras BE, Canizales-Quinteros S, Hernández-Reyna A, Mendoza-Vargas A, Ochoa-Leyva A. Gut dsDNA virome shows diversity and richness alterations associated with childhood obesity and metabolic syndrome. *iScience*. 2021 Aug 20;24(8):102900.
13. **Gallardo-Becerra L**, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, López-Contreras BE, Mendoza-Vargas A, Nielsen H, Ochoa-Leyva A. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. *Microb Cell Fact*. 2020 Dec;19(1):61.
14. Cornejo-Granados F, **Gallardo-Becerra L**, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. *PeerJ*. 2018 Aug 10;6:e5382.
15. Cornejo-Granados F, Lopez-Zavala AA, **Gallardo-Becerra L**, Mendoza-Vargas A, Sánchez F, Vichido R, Briebe LG, Viana MT, Sotelo-Mundo RR, Ochoa-Leyva A. Microbiome of Pacific Whiteleg shrimp reveals differential bacterial community composition between Wild, Aquacultured and AHPND/EMS outbreak conditions. *Sci Rep*. 2017 Dec;7(1):11783.